

Transfer Boundaries in Affective Audio Modelling: Zero-Shot Loss, Measured Adaptation Cost, and the Limits of the Available Evidence

Jingyi Zhang  and Xiaotong Yao

Abstract—Affective audio models are usually evaluated inside the corpus they were fitted on, and that optimism is rarely reported alongside the cost of removing it. Using four rated-sound corpora, five representations and three physiological corpora, we measure one mapping across four boundaries: to a new corpus, to edited audio, to a sensor in place of self-report, and to an individual listener. At each we report a within-corpus reference, what survives unadapted, and what adaptation recovered. Within a corpus, predicted arousal reaches 84% of what annotation reliability permits. An unadapted corpus swap costs about 15% of that reference within the music domain and 75% across it; in both cases 100 target labels recover roughly half the gap, label-free second-moment alignment recovers a quarter of the cross-domain gap, and four pretrained representations recover little of it. Editing audio moves model output, not listeners, and inconsistently across model and material. On the physiological boundary no acoustic route exceeds a random-projection control, and no fraction-of-ceiling is quotable: presentation position alone reproduces the target’s reliability. Per-person calibration loses to the population model at every budget existing corpora supply. We separate what improved, what did not under the routes tested, and what the evidence cannot decide.

Index Terms—Affective computing, cross-corpus generalisation, domain adaptation, few-shot adaptation, soundscape emotion, annotator agreement, personalisation, dataset bias.

I. INTRODUCTION

CHOOSING sound to shape affective state is an applied problem with a growing evidence base and a well-known reporting convention: accuracy is characterised within the collection on which the model was fitted. That convention answers none of the questions an application asks. An application changes the material, changes the *kind* of material, changes how the response is measured, and changes the listener. What matters is how much of the mapping survives each of those changes and what it costs to recover what is lost.

Both halves of that sentence are usually missing. Cross-corpus evaluations in affective audio are typically reported zero-shot: a model is fitted on one corpus, applied unchanged to another, and the resulting drop is reported as the generalisation gap. A drop measured that way is an upper bound on the problem, not an estimate of it, because no adaptation was attempted. The complementary error is to report an adaptation

method’s gain without a control that establishes how much of the gain is specific to the target rather than generic to any distribution correction.

This paper measures both. We push one mapping — group-averaged perceived arousal from acoustic descriptors — across four boundaries, ordered by how far each moves from the fitting conditions: across corpora, across the synthetic edits a deployed system applies to audio, across the level at which the response is measured, and across individuals. At each boundary we report three things: a within-corpus reference for that target, the fraction of it that survives an unadapted crossing, and what each adaptation route we tested recovered.

We separate three evidence states throughout, and label them explicitly rather than collapsing them into a verdict:

- 1) **Improvement observed.** A named adaptation route recovered a quantified fraction of the gap.
- 2) **No improvement observed under the routes tested.** The routes we ran did not recover the gap. This bounds those routes; it does not bound every route, and it is not evidence that the information required is absent from the input.
- 3) **Insufficient evidence to determine cause or recoverability.** The design cannot separate the competing explanations, and we say which.

This paper is a revision of a preprint [1] that reported the same experiments under a stronger claim: that generalisation failures divide into two kinds whose remedies are mutually exclusive, one of which no quantity of data can repair. That claim does not survive its own evidence, for a reason worth stating plainly because it is a general hazard. The learning curve that priced the recoverable gap was computed on the cross-domain corpus pairs, while the text attributed it to same-domain transfer and simultaneously described the cross-domain boundary as unpurchasable. The figure legend was correct and the prose was not. We report the corrected attribution here, withdraw the framework it supported, and record the full audit in the supplementary material.

Two findings of independent methodological value emerged from that audit, and both concern corpora in wide use. Half of Emo-Soundscapes is mixtures of thirty source recordings, and its distributed metadata gives every clip its own group identifier, so the standard grouped cross-validation that authors apply to it does not prevent near-duplicate clips from spanning folds (Section III-D). And in PMemo, stimulus-level reliability of electrodermal measures is reproduced — in one case exceeded — by presentation position alone, so those measures cannot be

J. Zhang is with Shanghai Huangan Technology Co., Ltd., Shanghai, China, and with Systems Neuroscience, Institute for Neuroscience, Department of Health Sciences and Technology (D-HEST), ETH Zurich, Zurich, Switzerland (e-mail: jingyi.zhang@hest.ethz.ch).

X. Yao is with Shanghai Huangan Technology Co., Ltd., Shanghai, China.

The authors are engaged in the development of a commercial sleep-audio product. See Section VII.

used as a stimulus-driven ceiling without a position control (Section IV-E). Anyone reusing either corpus is affected.

II. RELATED WORK

A. Cross-corpus affective audio prediction

That affective audio models degrade across corpora is established rather than novel. In speech, cross-corpus emotion recognition has been surveyed and benchmarked repeatedly, with normalisation strategies shown to account for a substantial share of the nominal gap [2], [3]. In music, cross-dataset evaluation reports similar degradation and traces part of it to differences in annotation protocol rather than in audio [4], and reviews note that most reported accuracies are within-corpus [5], [6]. Dataset bias in audio, as in vision, makes corpus identity itself predictable [7].

Our contribution is not the existence of the gap but its *pricing*. Where those studies report a zero-shot drop, we report the drop alongside what a named budget of target-side labels and what a label-free alignment recover, with a sham control that separates target-specific from generic gain. We also add a boundary those studies do not cross: between two sound domains that share no material, rather than between two corpora of the same kind.

B. Unsupervised and few-shot domain adaptation

Second-moment feature alignment (CORAL) [8] and subspace alignment [9] are the two standard closed-form, label-free baselines, and both are routinely reported without a control for what the whitening step alone achieves. Few-shot target-side supervision is the other standard route, and surveys of domain adaptation note that its budget is rarely quoted in units an application can act on [10]. Work on affective audio specifically has applied adversarial and discrepancy-based adaptation to cross-corpus speech emotion [11].

We add three things: the budget curve in labels rather than in epochs; a sham alignment that establishes how much of CORAL's gain is target-specific; and a decomposition of CORAL into per-dimension variance matching and correlation structure matching, which turns out to locate most of the gain in the second.

C. Personalisation and individual differences

That listeners differ in how acoustic properties map onto their reported affect is well documented, and personalised music emotion recognition has been proposed repeatedly [12]–[14]. The question an application faces is narrower and less often asked: at what observation count per person does a personal model beat the population model. Work on the statistics of individual differences warns that between-person variance components estimated from few observations per person are biased upward [15].

Our contribution is to estimate that break-even point on three corpora with a common estimator and a per-cell null, and — crucially — to test it out of sample. The out-of-sample test is the part that changes the conclusion: the variance model places break-even at 67 to 318 observations per person, and a direct held-out comparison returns a negative personalisation gain at every budget the corpora actually supply.

D. Annotator agreement as a ceiling

Correcting an observed correlation for the unreliability of its target is standard psychometrics [16], [17], and affective computing has used inter-annotator agreement to bound achievable performance [18]. Two errors are common and we guard against both: dividing by the reliability rather than by its square root, which inflates the corrected value by a factor of $1/\sqrt{r}$; and pairing a correlation measured on one corpus with a reliability estimated on another.

We add the observation that a reliability estimate is only a ceiling if it is stimulus-driven, and that this has to be tested. On the electrodermal measures of PMemo it fails: position alone reproduces the reliability, so no ceiling can be quoted.

III. DATA AND METHODS

A. Corpora

Four rating corpora were used: DEAM (music, $n = 1,233$) [19], PMemo (music, $n = 717$) [20], Soundtracks Set 1 (music, $n = 360$) [21] and Emo-Soundscapes (environmental, $n = 1,213$) [22]. ESC-50 [23] served as a second environmental corpus for the category control. The prediction target throughout is group-averaged perceived arousal, rescaled to $[-1, 1]$; the circumplex framing follows [24], [25]. Soundtracks additionally carries seven scales (valence, tension, anger, fear, happiness, sadness, tenderness) on nine-point Likert scales, linearly rescaled — a monotone transformation that leaves all reported rank statistics unchanged.

Soundtracks was included to test whether within-music transfer reflects shared acoustic structure or a shared annotation tradition: it originates from a different laboratory and decade, uses a categorical rather than continuous protocol, and was assembled as a balanced design of twelve target emotions $\times$ 30 excerpts.

One asymmetry between the music and environmental corpora must be recorded here because it limits every cross-domain statement in this paper. Emo-Soundscapes was annotated by *pairwise comparison* and its ratings derive from a ranking procedure, whereas the three music corpora were rated directly on continuous or Likert scales. The corpora therefore differ in annotation method as well as in sound domain, and with one environmental corpus available the two cannot be separated (Section V-C).

B. Audio preparation and descriptors

All audio was resampled to 22,050 Hz, centre-cropped to at most 30 s, and loudness-normalised to -23 LUFS (ITU-R BS.1770) before any descriptor was computed. Normalisation is not cosmetic: without it, corpus-specific mastering levels act as an identifier that inflates apparent cross-corpus transfer (Supplementary Fig. S1). For the duration control, music excerpts were additionally cropped to a 6 s centred window to match the environmental corpus.

Spectro-temporal (122). Mel-frequency cepstral coefficients, spectral centroid, bandwidth, contrast, flatness and roll-off, zero-crossing rate, root-mean-square level, onset strength and onset rate, spectral slope, and band-energy ratios. Each frame-level

series is summarised by mean, standard deviation, 10th, 50th and 90th percentiles, and the mean absolute first difference, the last of which indexes rate of change rather than level. Computed with librosa [26].

Tonal (39). A key-invariant chroma profile obtained by estimating the tonic through correlation with Krumhansl–Kessler profiles [27] and rotating so the tonic occupies index 0; major- and minor-key strength and their difference (mode score); chroma flux and entropy; tonal centroids; an interval-consonance measure; sensory roughness after Plomp and Levelt [28]; and fundamental-frequency statistics. The full list is in Supplementary Section S11. Rotation to the tonic is essential rather than convenient: an unrotated chroma vector encodes which key a piece is in, which is arbitrary with respect to affect, so an unrotated representation would let a model learn a corpus’s key distribution. The consonance weights are fixed a priori and are not fitted.

Pretrained representations. Four families spanning four pretraining paradigms were extracted over all four corpora: AST, supervised classification on AudioSet [29], [30]; MERT, self-supervised on music [31]; Wav2Vec2, trained on speech [32]; and CLAP, audio–text contrastive learning [33]. CLAP’s pre-training set includes Freesound, from which Emo-Soundscapes was built. We have not performed a sample-level overlap check and therefore describe the overlap as *possible at the corpus level* rather than verified, and report CLAP separately throughout. The Wav2Vec2 checkpoint used is a speech model; we do not claim it is purely self-supervised, only that its pretraining domain is speech.

C. Models and evaluation

Six regressors — ElasticNet, support vector regression, k -nearest neighbours, random forest, gradient boosting and XGBoost [34] — were evaluated by five-fold GroupKFold cross-validation. Cross-corpus transfer fits on the whole of one corpus and evaluates on the whole of another, with nothing refitted. Reported values are Spearman rank correlations. Where a single value is reported for a cell it is the median over the six algorithms unless stated otherwise; the preprint reported the best of six in several places, which carries a winner’s bias, and we now report the best only where the quantity is explicitly an upper bound. Where a comparison between conditions is made, all six are retained as paired observations and the comparison is tested by Wilcoxon signed-rank.

Annotation reliability. Per-rater judgements were split at random into two halves, stimulus means computed within each, correlated, and corrected by the Spearman–Brown formula [16]; the reported value is the mean over 200 random splits. Where a model’s performance is expressed as a fraction of what the annotation permits, the divisor is $\sqrt{r}$, not r : the maximum correlation an errorless predictor can achieve against a target of reliability r is $\sqrt{r}$ [17]. Dividing by r would inflate the figure by $1/\sqrt{r}$.

D. Two corpus-level hazards found during this revision

1) *Emo-Soundscapes contains mixtures of a small source set*: Of the 1,213 Emo-Soundscapes clips, 600 are single Freesound

TABLE I
EFFECT OF EXCLUDING THE 613 EMO-SOUNDSCAPES MIXTURE CLIPS.
MEDIAN OVER SIX ALGORITHMS; TRANSFER MEDIAN OVER THE SIX
ORDERED CORPUS PAIRS OF EACH KIND $\times$ SIX ALGORITHMS. LOSS IS PER
PAIR AGAINST THE TARGET CORPUS’S OWN WITHIN-CORPUS REFERENCE,
THEN MEDIAN OVER PAIRS.

Quantity	All 1,213	600 originals
Within-corpus ρ	0.744	0.646
Cross-domain ρ , median	0.043	0.129
Cross-domain ρ , mean (s.d.)	0.058 (0.16)	0.115 (0.15)
Cross-domain loss	93.3%	74.9%
Same-domain loss	15.0%	15.0%

recordings and 613 are mixtures. The mixtures are built from five selected clips in each of the six Schafer categories — **thirty source recordings in total** — combined two or three at a time at two relative levels each. The distributed filenames encode the constituents, from which we recover that the 613 mixtures occupy only **161 distinct constituent sets**, a mean of 3.81 clips per set and up to 8. Clips in one set differ only in mixing level.

The label table gives every clip its own group identifier, so GroupKFold on this corpus is arithmetically identical to KFold: near-duplicate mixtures span folds, and a training clip can be a constituent of a test mixture. The protocol we and others describe as preventing recording-level leakage does not prevent it here.

We therefore report every Emo-Soundscapes quantity twice: on all 1,213 clips (*published*, for comparability with prior work including our own preprint) and on the 600 originals alone (*clean*), where every clip is a distinct recording and construction leakage is impossible. The effect is substantial in both directions and is given in Table I: the within-corpus reference falls from 0.744 to 0.646, and median cross-domain transfer *rises* from 0.043 to 0.129, because a test set half composed of near-duplicates distorts the ranking the metric is computed on. Both changes shrink the apparent domain gap. The clean variant is the one we interpret.

2) *PMemo electrodermal reliability is reproduced by presentation position*: Stimulus-level reliability of the three PMemo electrodermal measures is 0.116 (skin-conductance response rate), 0.237 (phasic mean) and 0.234 (skin-conductance level slope). A sham that uses *presentation position alone* as the stimulus identifier returns 0.290, 0.283 and 0.267 — that is, 249%, 120% and 114% of the values obtained with the true stimulus labels. The assignment of songs to positions in this corpus is itself significantly non-random (intraclass correlation 0.041, $p = 0.005$), and regressing position out of the measures reduces their reliability to 0.038, 0.081 and 0.206.

The consequence is not that the electrodermal measures are meaningless but that their stimulus-level reliability is not established as stimulus-driven, so it cannot serve as a ceiling. The preprint expressed the cross-response-channel result as a percentage of exactly that ceiling. That percentage is withdrawn (Section IV-E).

E. Adaptation routes

Four label-free routes and one supervised route were compared, expressed as the fraction of the gap between zero-shot transfer and a model trained on the target corpus itself. Label-free: per-corpus z -scoring; CORAL [8], matching second moments in closed form; diagonal CORAL, matching per-dimension variance only; and subspace alignment [9]. Supervised: *few-shot*, adding $k \in \{10, 25, 50, 100, 200\}$ randomly chosen target-corpus examples to the training set and evaluating on the remaining target examples, averaged over 20 resamples. The estimator for all adaptation results is ridge regression with a cross-validated penalty, chosen on training data only.

Two reporting points follow from the audit. First, the few-shot numerator is ridge while the denominator in the preprint was a six-algorithm median; we now give both denominators. Second, $k = 200$ is unavailable for Soundtracks ($n = 360$, which the protocol's $k < n/2$ rule excludes), so the pair set at $k = 200$ differs from the pair set at smaller k . We therefore report the curve both over all available pairs and over the pairs present at every k .

Sham alignment. CORAL performs two operations at once: it whitens the source and it recolours towards the target. To separate them, the source was also aligned to an unrelated third corpus, which by construction carries no target-specific information. With a single environmental corpus this control is only valid on half the pairs: when the source is environmental, the third corpus is necessarily music and therefore in the *target's* domain. We report the sham separately for the three valid and three invalid pairs. The preprint pooled them, which inverted the conclusion.

F. Physiological corpora and controls

EEG. OpenNeuro ds002721 [35]: 31 listeners, 19 scalp channels at 1 kHz referenced to FCz, six runs each, four containing ten 12 s music excerpts. Its stimuli are drawn from the same published stimulus set as our fourth music corpus: the dataset's README gives their source as [21], from which Soundtracks Set 1 also comes. The two are therefore not independent material, and we have not been able to establish from the distributed metadata which 40 of those excerpts were used. Event annotation follows BIDS and its EEG extension [36], [37]. Processing used MNE-Python [38]: 50 Hz notch, 1–45 Hz bandpass, resampling to 250 Hz. Spectral and asymmetry measures were computed after average re-referencing; coherence was computed on the recording reference, because average referencing introduces a common component that inflates coherence.

Trial-wise measures (156): relative band power in five bands $\times$ five regions as change from a 2 s pre-onset baseline to a 1–14 s window (26); magnitude-squared coherence between all ten region pairs in five bands (50); the imaginary part of coherency for the same pairs and bands (50), insensitive to zero-lag volume conduction; and log-ratio asymmetry for six homologous electrode pairs in five bands (30). Coherence and asymmetry are included because the original analysis of these recordings located its effect there [39]; restricting the set to

band power would test the conclusion on a representation that excludes the effect in question.

Four dataset-level controls were run before any substantive analysis, with the rule that failure of any one halts interpretation: event bookkeeping; occipital $>$ frontal relative alpha at rest; event-to-sample alignment via independently annotated blink events; and a stimulus-locked evoked response against a subject-wise sign-flip null with max-statistic correction [40]. An individual-level control required each listener's EEG to predict trial index, which is necessarily predictable given impedance and vigilance drift. Multiplicity was controlled by Benjamini–Hochberg [41].

Electrodermal. The PMemo corpus carries song-level electrodermal recordings, which we use for the acoustics-to-physiology comparison. This pipeline was not described in the preprint and was not positive-controlled; both are corrected here. Processing used NeuroKit2 [42] following standard recommendations [43], with phasic–tonic decomposition. Because no positive control for this pipeline passed — and because of the position confound above — results from it are reported as raw correlations against an explicit random-projection reference, and not as a fraction of any ceiling.

A second electrodermal corpus, BIRAFFE2 [44], was analysed and withdrawn: group-level habituation ran opposite to the established direction, post-stimulus windows did not exceed random windows, and applied per participant, none of 94 passed both controls (Supplementary Section S2). It supports no conclusion in this paper, including by way of corroboration.

Measured design requirement. OpenNeuro ds003690 [45]–[47]: 75 adults (36 aged 19–30, 39 aged 50–70), 64-channel EEG, bipolar electrocardiogram and binocular pupillometry at 500 Hz during cued reaction-time tasks. Auditory cues were 250 ms pure tones at approximately 67 dB(A); each participant contributed 240 cues across two active tasks and 30 in a passive block. One response measure was fixed per channel before analysis — a fronto-central window mean for EEG, a heart-rate change and a pupil-diameter change, each against its own pre-cue baseline; exact windows and rejection criteria are in Supplementary Section S11. Each real cue was paired with a randomly placed anchor in the same recording and all analyses operate on the difference, under which sign-flipping is exact. Detection was assessed per participant with 5,000 sign flips at $\alpha = 0.05$ with Benjamini–Hochberg correction.

G. Per-person calibration

For each descriptor–outcome cell the between-person standard deviation τ of the acoustic slope was estimated by restricted maximum likelihood and compared with a null built by parametric bootstrap from that cell's own design, because τ estimated from few observations per person is biased upward: at 18 observations per person the estimator returns 0.11 when the true value is zero. Break-even n^* is the number of observations per person at which the corrected spread exceeds the error in estimating it.

n^* is a property of a variance model, not a demonstration that personalisation pays. We therefore also report the direct test,

which the preprint omitted: fitting each listener their own slope and comparing out-of-sample prediction against the population slope, at budgets $n \in \{18, 30, 50, 100\}$.

Three settings were used, and they differ in more than the quantity under study. Urban soundscape ratings come from ARAUS [48], [49] with ISO-532 psychoacoustic descriptors (loudness, sharpness, roughness, fluctuation strength) against the pleasantness and eventfulness axes; music ratings come from DEAM with five spectro-temporal and tonal descriptors against arousal and valence; and electrodermal responses come from PMemo with the same five descriptors against three electrodermal measures. Corpus, descriptor set and outcome axis vary together across the three settings, so differences between them cannot be attributed to any one of the three (Section V-C).

H. On claims of prior specification

The preprint described several criteria as fixed in advance, pre-declared, or falsifiable in advance. We have been unable to produce a traceable artifact predating the corresponding analyses for any of them: the manuscript directory was untracked in version control, the repository head predates most of the runs, and only 2 of 199 run records carry a declared hypothesis. Those phrasings are removed throughout and no analysis in this paper is presented as preregistered. Where a criterion genuinely constrained a decision we say what the criterion was and leave the reader to weight it accordingly.

IV. RESULTS

A. Within-corpus reference and the annotation ceiling

Across the four corpora, models predicting group-averaged perceived arousal from 122 spectro-temporal descriptors reach Spearman $\rho = 0.65$ – 0.81 under grouped five-fold cross-validation (medians over six algorithms: DEAM 0.724, PMemo 0.801, Soundtracks 0.810, Emo-Soundscapes (600 originals) 0.646).

Converting that into a distance from what the annotation permits requires the model performance and the annotation reliability to come from the same excerpts. On the 717 PMemo excerpts, with the same descriptors, the same folds and a per-rater split-half reliability of 0.935, the best of the six algorithms reaches $\rho = 0.817$. Dividing by $\sqrt{0.935}$ puts it at **84.4%** (95% CI 81.5–87.0%) of the correlation an errorless predictor of these annotations could attain.

Three qualifications belong with that figure, because it is optimistic in three separate ways. It is the best of six algorithms evaluated on one set of out-of-fold predictions, and the selection is not itself cross-validated. The bootstrap resamples excerpts but does not refit, so it expresses sampling variability in the evaluation and not in the fitting. And the reliability estimate is carried through as fixed rather than as a quantity with its own interval. The value should be read as an upper bound.

It should also not be read as a share of variance: the attenuation-corrected value is a ratio of correlations, and squaring it puts the models at about 71% of the reproducible variance, leaving roughly 29% that better annotation would not explain away. What the figure establishes is narrow and

sufficient: within a corpus, the remaining headroom for a better model is small relative to the losses measured below.

Two framing choices affect the reference rather than the model. Treating arousal as continuous rather than as a high-versus-low contrast retains the middle of the range, which a median split discards along with a third to a half of the excerpts (Supplementary Fig. S2). And an additive glass-box model, whose per-descriptor shape functions can be read directly, costs almost nothing in accuracy relative to the tree ensembles (Supplementary Figs. S3, S4).

B. Boundary 1: cross-corpus

1) *What an unadapted swap costs:* Transfer between corpora of the same domain is largely preserved. Over the six ordered music-to-music pairs, each evaluated with the six algorithms (36 values in total, from 6 pairs — the preprint described these as 36 pairs, which overstated the number of independent units six-fold), transfer averages $\rho = 0.591$ (s.d. 0.131). Expressed per pair against the target corpus's own within-corpus reference, the median loss is **15.0%** (range 10.7–35.6%).

Between music and environmental sound, over the six ordered cross-domain pairs $\times$ six algorithms, transfer averages $\rho = 0.115$ (s.d. 0.153) on the clean environmental corpus, a median per-pair loss of **74.9%** (range 58.6–115.6%). On the published corpus the same quantities are 0.058 and 93.3%; the difference is the mixture leakage of Section III-D, and it accounts for roughly two-thirds of the gap between "four-fifths of performance lost" and "all of it" (Fig. 1).

Three controls exclude the obvious confounds. Truncating music excerpts from 30s to a 6s window, matching the environmental corpus, leaves same-domain transfer at 0.63 and cross-domain transfer near zero. Holding out each of the six Schafer categories in turn leaves generalisation within the environmental corpus at $\rho = 0.52$, so the collapse is not simply a response to unseen content. Loudness normalisation before feature extraction removes a corpus fingerprint that otherwise inflates apparent transfer (Supplementary Fig. S1).

The barrier is not symmetric. Transfer from the environmental corpus into each of the three music corpora is uniformly low ($\rho \leq 0.09$), while transfer outward is graded: -0.04 from PMemo, $+0.13$ from DEAM, $+0.24$ from Soundtracks. Film soundtracks contain substantial atmospheric material, which is a plausible account of that ordering but not one we measured; we report the asymmetry and not an explanation for it.

2) *What adaptation recovers:* Calling a loss recoverable is only meaningful if the budget is quoted. Fig. 2 gives the recovery curves, and Table II the numbers.

Target-side labels recover a substantial fraction of both gaps. On the cross-domain pairs, 100 labels recover 57% of the gap and 200 recover 71%; on the same-domain pairs, 100 recover 50% and 200 recover 77%. The two curves differ in shape rather than in whether a price exists: the cross-domain curve rises faster at small budgets, because its zero-shot starting point is much lower, while the same-domain curve overtakes it at 200. Restricting both to the pairs present at every budget changes the cross-domain values by at most 2 points and the same-domain values by up to 5, and does not change the ordering.

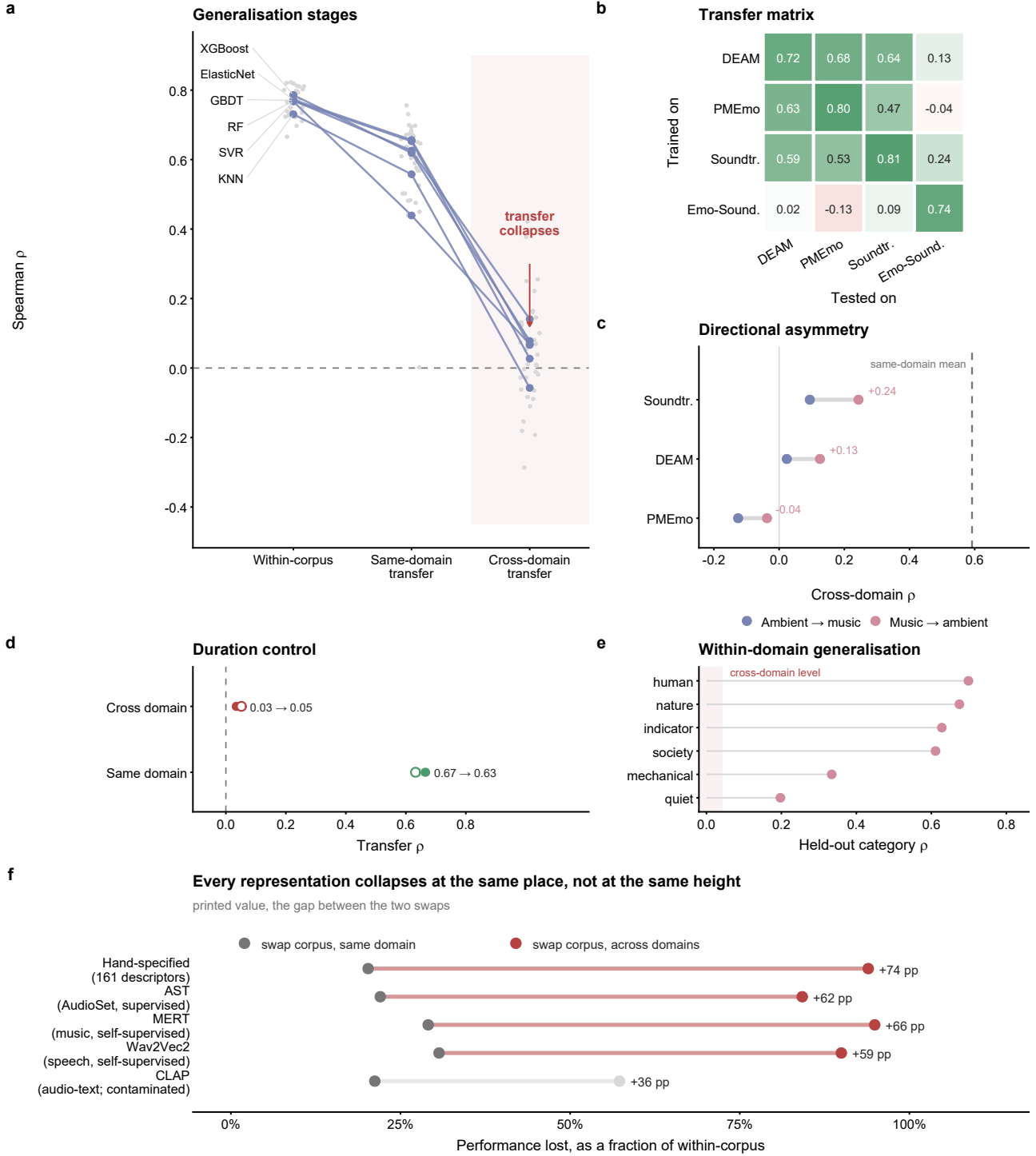


Fig. 1. Arousal models transfer between corpora of the same sound domain far better than across it, under every representation tested. **(a)** Generalisation stages for six algorithms over 122 spectro-temporal descriptors, from within-corpus cross-validation through same-domain transfer to cross-domain transfer; grey points are the individual corpus pairs. **(b)** Transfer matrix averaged over algorithms; the diagonal is within-corpus cross-validation. **(c)** The barrier is asymmetric: inward transfer from the environmental corpus is uniformly low ($\rho \leq 0.09$) while outward transfer is graded with the source corpus. **(d)** Duration control (30 s filled, 6 s open). **(e)** Leave-one-category-out generalisation within the environmental corpus. **(f)** The same corpus swap costed under five representations, as the fraction of each representation's own within-corpus performance lost. Values in **(a)**–**(f)** are on the published environmental corpus, for comparability with prior work; Table I gives the leak-free counterparts. Metric is Spearman ρ ; all audio loudness-normalised to -23 LUFS. Extended description and source data in the Supplementary Information.

This is the correction that matters most in this revision. The preprint reported the cross-domain curve — 20/34/50/64/69% at $k = 10/25/50/100/200$ — and attributed

it in the abstract and discussion to transfer "to a new corpus of the same kind", while describing the cross-domain boundary as one on which no purchase could be found. Both halves were

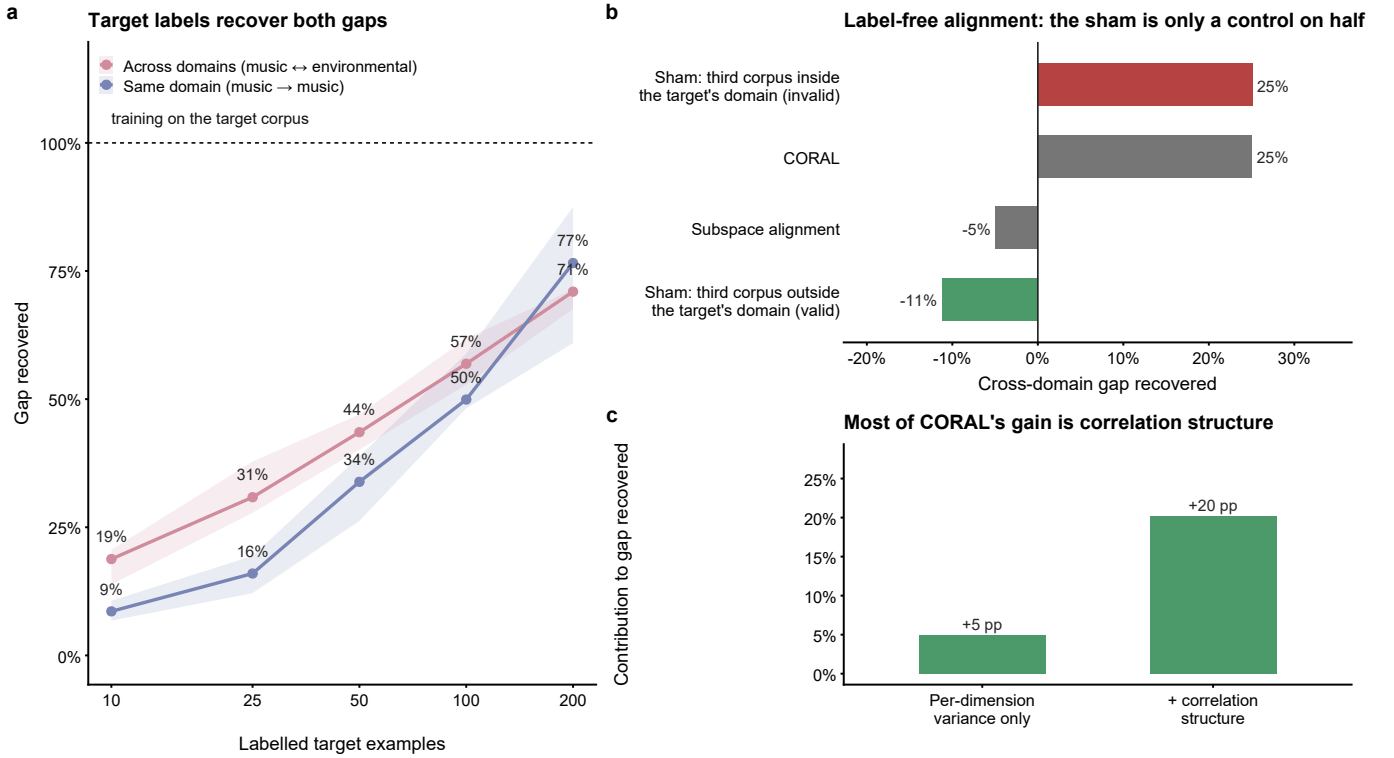


Fig. 2. What adaptation recovers on the corpus boundary, as the fraction of the gap between zero-shot transfer and a model trained on the target corpus itself. Environmental corpus without its mixture clips throughout. (a) Budget curves for both kinds of corpus swap; line and points, median over the six ordered pairs of each kind; band, interquartile range. *The preprint drew only the cross-domain curve and reported it as the same-domain price.* (b) Label-free alignment with its sham control split by validity: with one environmental corpus the “unrelated third corpus” lies inside the target’s own domain on three of the six pairs, and the entire pooled sham effect comes from those three. (c) CORAL decomposed; matching correlation structure, which is target-specific by construction, contributes 80% of the gain. Ridge estimator; metric is Spearman ρ .

wrong. The curve is the cross-domain curve, so the boundary the preprint called unpurchasable is precisely the one whose price it quoted; and the same-domain price, which the preprint’s numbers were said to describe, is different.

A label-free route also works. Correlation alignment recovers **25%** of the cross-domain gap with no target labels at all. The preprint discounted this to roughly a third of its face value on the strength of a sham control that appeared to recover 14%. That pooled figure is an average over three pairs where the sham is a valid negative control and three where it is not: when the source corpus is environmental, the “unrelated third corpus” is necessarily music and therefore in the target’s domain. Separated, the valid sham recovers -11% and the invalid sham $+25\%$. The entire apparent sham effect comes from the invalid half, and CORAL’s gain is therefore *more* target-specific than the preprint claimed, not less. Decomposing CORAL confirms this: matching per-dimension variance alone recovers 5%, and the further 20 points — 80% of the total — come from matching correlation structure, which is target-specific by construction. Subspace alignment is worse than doing nothing (-5%).

One claim the preprint made about these curves does not survive either. It reported that the return per doubling of the budget collapses after 100 labels, from a constant 14–16 points to 5. On the clean environmental corpus the increments are 10, 14, 10 and 14 points — no collapse, and the largest increment

TABLE II
ADAPTATION ON THE CORPUS BOUNDARY, AS THE FRACTION OF THE GAP BETWEEN ZERO-SHOT TRANSFER AND A MODEL TRAINED ON THE TARGET CORPUS ITSELF. RIDGE ESTIMATOR, CLEAN ENVIRONMENTAL CORPUS, MEDIAN OVER THE SIX ORDERED PAIRS OF EACH KIND. TWO DENOMINATORS ARE GIVEN FOR THE FEW-SHOT ROWS BECAUSE THE NUMERATOR IS RIDGE WHILE THE SIX-ALGORITHM MEDIAN IS THE REFERENCE USED ELSEWHERE IN THIS PAPER.

Route	Same domain		Across domains	
	6-algo ref.	ridge ref.	6-algo ref.	ridge ref.
10 target labels	8.6%	7.5%	18.8%	17.9%
25 target labels	16.0%	13.7%	30.9%	28.7%
50 target labels	33.9%	28.1%	43.5%	41.3%
100 target labels	49.9%	43.6%	56.9%	54.3%
200 target labels	76.5%	60.6%	71.0%	65.3%
CORAL (no labels)	—	—	25.0%	—
diagonal only	—	—	4.9%	—
sham, valid pairs	—	—	-11.1%	—
sham, invalid pairs	—	—	25.1%	—
Subspace alignment	—	—	-5%	—

is the last. The reported collapse was produced jointly by the mixture leakage and by the change of pair set at $k = 200$. We therefore make no claim about where the curve saturates: it had not saturated at the largest budget we tested.

3) *Representation choice*: The most direct objection to the preceding is that 122 hand-specified descriptors are too weak and that a representation learnt from a large audio corpus would cross what they cannot. We extracted four pretrained representations spanning four pretraining paradigms and repeated the transfer analysis unchanged (Table III).

The representations transfer between corpora and they do not cross between domains. For a given representation, the same operation — swap the corpus, refit nothing — costs 18–23% of within-corpus performance when the new corpus is in the same domain and 79–108% when it is not. That disposes of the alternative reading in which the embeddings simply fail to move between datasets at all.

AST is the case that matters most, because AudioSet contains large numbers of environmental categories and it is therefore the family that most ought to cross. Here the mixture-leakage correction changes the conclusion quantitatively. On the published environmental corpus, AST raises cross-domain transfer from the hand-specified descriptors' $\rho = 0.047$ to 0.125, a factor of 2.7 — the figure the preprint reported. On the leak-free corpus the same comparison is 0.129 against 0.162: a difference of +0.034 and a factor of **1.26**. Most of the apparent advantage of a large pretrained representation over 122 hand-specified descriptors at this boundary was an artifact of evaluating on a test set half composed of near-duplicate mixtures. What remains is a small improvement in the expected direction.

Wav2Vec2, pretrained on speech, transfers across domains at $\rho = -0.034$ on the leak-free corpus, which is to say not at all. MERT, pretrained on music, reaches 0.085.

CLAP is the exception at a loss of 56%, and it is the family whose pretraining set includes the repository from which the environmental corpus was built. It is not a generally stronger representation: within a corpus it reaches 0.774 against AST's 0.770, and on music-to-music transfer — the comparison that excludes the environmental corpus entirely — 0.620 against AST's 0.619 and 0.614 for the hand-specified descriptors. Its advantage appears only in comparisons that necessarily involve the corpus its pretraining data may overlap. We have not run a sample-level overlap check, so we cannot distinguish genuine crossing from familiarity with the target, and with one environmental corpus available no comparison in this paper can. CLAP is reported and flagged and is used neither as support nor as refutation.

The statement this table supports is therefore: *under the four pretraining paradigms tested, changing the representation does not recover the cross-domain gap, and the best of them improves on hand-specified descriptors by 0.034 in rank correlation*. It is not a statement that no representation could.

C. Representation choice is target-specific: perceived happiness

One result shows what a representation change does buy, and it is the clearest demonstration in this paper that representation choice is target-specific rather than uniformly good or bad.

On Soundtracks, which carries eight rating scales, acoustic descriptors resolve tension ($\rho = 0.804$), tenderness (0.754), valence (0.735), fear (0.717) and anger (0.715). Perceived happiness is the single outlier at 0.474 (Fig. 3a).

TABLE III

TRANSFER UNDER FIVE REPRESENTATIONS, EACH EVALUATED WITH THE SAME MODELS, FOLDS AND SCORING. MEDIAN OVER THE RELEVANT CORPUS PAIRS $\times$ SIX ALGORITHMS. *Within corpus* IS CROSS-VALIDATED PERFORMANCE ON A CORPUS'S OWN DATA; *same domain* IS TRANSFER TO A DIFFERENT CORPUS OF THE SAME KIND; *across domains* IS TRANSFER BETWEEN MUSIC AND ENVIRONMENTAL SOUND. THE UPPER BLOCK USES THE ENVIRONMENTAL CORPUS AS DISTRIBUTED; THE LOWER BLOCK EXCLUDES ITS 613 MIXTURE CLIPS (SECTION III-D). ONLY THE LOWER BLOCK IS INTERPRETED. CLAP'S PRETRAINING DATA INCLUDE THE SOURCE OF THE ENVIRONMENTAL CORPUS; IT IS REPORTED FOR COMPLETENESS AND EXCLUDED FROM THE CLAIM.

Representation	Spearman ρ			Loss from swap	
	Within corpus	Same domain	Across domains	Same domain	Across domains
<i>Environmental corpus as distributed (1,213 clips)</i>					
Hand-specified (122)	0.769	0.614	0.047	20.2%	93.9%
AST (AudioSet)	0.794	0.619	0.125	22.0%	84.2%
MERT (music)	0.737	0.522	0.038	29.1%	94.9%
Wav2Vec2 (speech)	0.511	0.354	0.051	30.7%	90.0%
CLAP (overlap)	0.787	0.620	0.336	21.2%	57.3%
<i>Environmental corpus without mixtures (600 clips)</i>					
Hand-specified (122)	0.751	0.614	0.129	18.3%	82.9%
AST (AudioSet)	0.770	0.619	0.162	19.5%	78.9%
MERT (music)	0.679	0.522	0.085	23.0%	87.4%
Wav2Vec2 (speech)	0.460	0.354	-0.034	23.0%	107.5%
CLAP (overlap)	0.774	0.620	0.342	19.9%	55.8%

Musical happiness depends on mode and harmony, which the spectro-temporal set does not represent. Adding the 39 tonal descriptors raises happiness by +0.132 against a mean of +0.045 for the other seven axes, and the ordering of gains follows mode-dependence: happiness, valence (+0.083) and sadness (+0.077) lead, while energy (+0.012) and anger (+0.025), which depend on level, roughness and rhythm, gain least. The deficit was representational rather than dimensional: 39 tonal descriptors *alone* reach $\rho = 0.580$ for happiness, above the 122 spectro-temporal descriptors at 0.474 (Fig. 3b). A mechanism check dissociates the two families: among all 161 descriptors, major-key strength and mode score rank 1st and 2nd for happiness but 3rd and 20th for tension, whereas interval consonance and sensory roughness rank 4th and 7th for tension but 117th and 76th for happiness (Fig. 3c).

The preprint presented this as a falsification test of its two-kind framework: an information-limited gap that a representation closes. We withdraw that framing for two reasons. The framework is withdrawn, so there is nothing to falsify; and the test's necessary first half — that additional data does *not* close the happiness gap — was never measured. No sample-size curve exists for any Soundtracks axis. What the result supports is the narrower and still useful statement that a targeted representation change can recover a specific deficit that more of the same descriptors would not obviously recover, and that which representation helps depends on which axis is being predicted.

Tonality is not the mechanism of the domain gap. Adding tonal descriptors *lowers* music-to-environmental transfer by 0.021 (18 paired comparisons, $p = 0.038$ uncorrected), in the predicted direction but at a magnitude accounting for roughly 4% of the gap. Tonal descriptors contribute almost nothing to

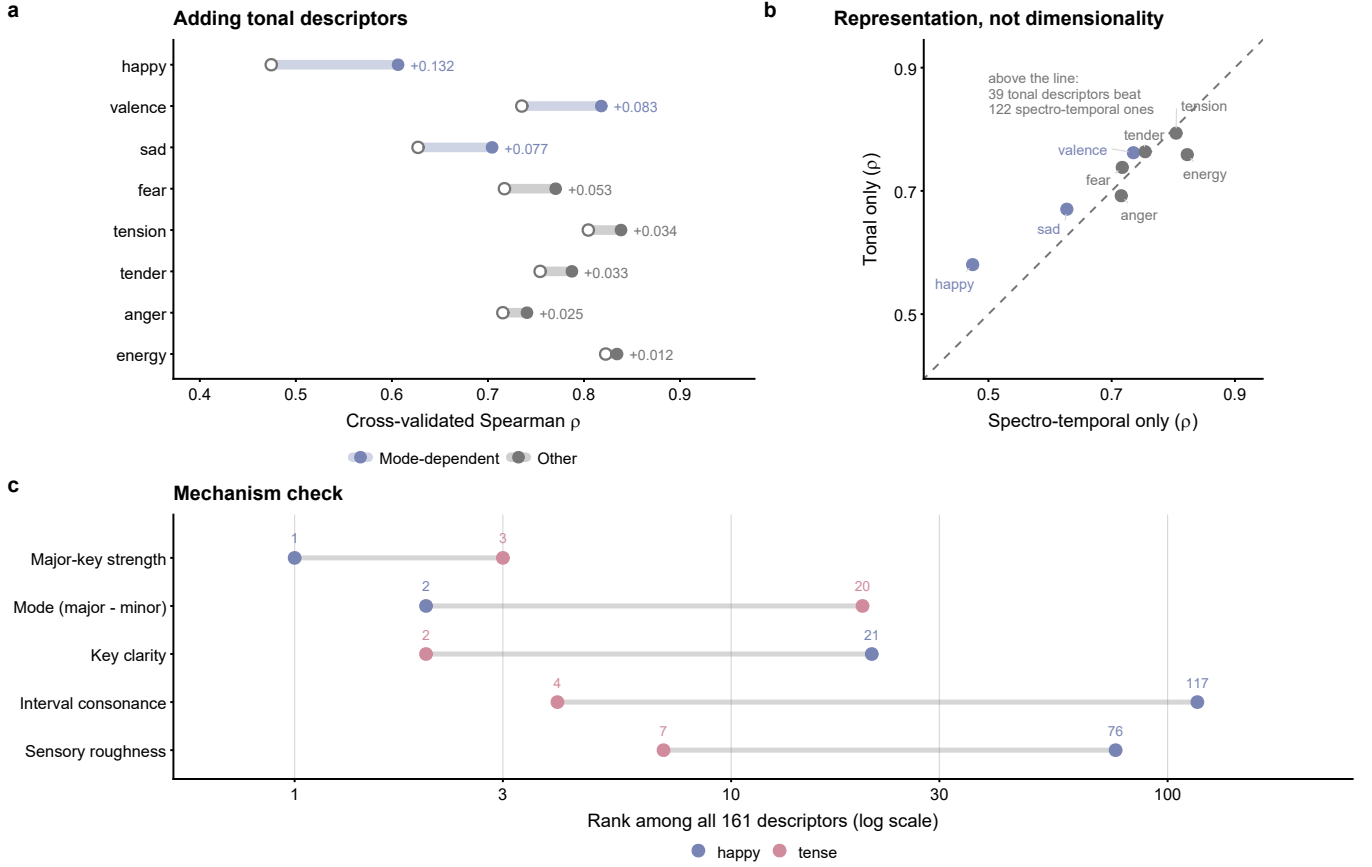


Fig. 3. A representation change recovers a specific deficit. (a) Cross-validated Spearman ρ for the eight affect axes of Soundtracks ($n = 360$) with 122 spectro-temporal descriptors alone (open) and with 39 tonal descriptors added (filled); axes ordered by gain. Blue marks the three axes for which musical mode is theoretically decisive, a grouping fixed before the comparison. (b) Tonal descriptors alone against spectro-temporal alone: for happiness, 39 tonal descriptors ($\rho = 0.580$) outperform 122 spectro-temporal ones (0.474), so the deficit was representational rather than dimensional. (c) Mechanism check: rank of five tonal descriptors among all 161 for happiness and for tension. Mode carries perceived happiness; consonance and roughness carry perceived tension. The preprint presented this as a falsification test of a two-kind framework; that framing is withdrawn, since the framework is withdrawn and the test's other half — that more data does not close the happiness gap — was never measured.

arousal specifically (+0.006 within the music domain), and arousal is the target on which the gap was measured.

D. Boundary 2: cross-synthesis

A deployed system never plays the corpus; it plays audio that has been filtered, compressed, layered or generated. To ask whether the descriptors a model relies on are ones that *move* a prediction when altered, we edited the audio: 80 source excerpts (40 music, 40 environmental) $\times$ three interventions $\times$ four doses, with the model held fixed and only the input changed (Fig. 4).

The perturbed models are binary random-forest classifiers of low- versus high-arousal, one fitted on each domain; the modelled quantity is the predicted probability of the low-arousal class, so a positive dose slope means the edit made the excerpt read as *calmer*. These are not the regression models of Section IV-B, a distinction the preprint did not make.

Two of the three interventions passed their manipulation check — the edit moved the descriptor it was designed to move, monotonically across dose. Dose response was tested per excerpt: a slope across the four doses, and a Wilcoxon signed-rank test of those slopes against zero.

1) *The direction of the effect is not consistent across model and material:* The preprint reported that spectral smoothing raised the low-arousal probability in both domains (median slope +0.015, sign consistency 70% for the music model; +0.074, 71% for the environmental model) and concluded that "the sign of an acoustic manipulation crosses the boundary that the predictions cannot, so what fails is calibration rather than the mapping".

Those statistics pool, for each model, the 40 excerpts of its own domain with the 40 of the other. Disaggregated, the pattern does not support the conclusion (Table IV). The music model shows the effect on both kinds of material. The environmental model shows it strongly on *music* excerpts (median +0.080, 100% sign consistency) and *not* on environmental excerpts — the material it was fitted on and the material the claim is about — where the median slope is -0.019 in the wrong direction at 57.5% sign consistency and $p = 0.073$.

We therefore withdraw the claim. Three of the four model $\times$ material cells move in the intended direction and one does not, and the one that does not is the cell the mechanism claim needs most.

A second defect affects how the sign-consistency statistic

TABLE IV

SPECTRAL SMOOTHING, DISAGGREGATED BY MODEL AND BY THE DOMAIN OF THE MATERIAL. PER-EXCERPT DOSE SLOPES ON THE PREDICTED LOW-AROUSAL PROBABILITY; POSITIVE MEANS THE EDIT MADE THE EXCERPT READ AS CALMER. THE POOLED ROWS ARE THE STATISTICS THE PREPRINT REPORTED.

Model	Material	n	Median slope	Wilcoxon p
Music	pooled (as reported)	80	+0.015	3×10^{-4}
	music excerpts	40	+0.099	2×10^{-4}
	environmental	40	+0.012	7×10^{-4}
Environmental	pooled (as reported)	80	+0.074	$< 10^{-4}$
	music excerpts	40	+0.080	$< 10^{-4}$
	environmental	40	-0.019	0.073

should be read. It is defined as the fraction of excerpts whose slope matches the sign of the observed median, so 50% is its arithmetic floor rather than a chance level; under the null its median is 0.54. Values near 50% therefore indicate no per-excerpt effect, and values of 57–60% are close to that floor. Only the cells at 70% or above are meaningfully above it.

Independently of the data, the mechanism claim was incompatible with the metric. Every transfer number in this paper is a Spearman rank correlation, which is exactly invariant under any strictly monotonic recalibration of a model's output. A purely calibrational deficit therefore *cannot* move ρ at all, let alone from 0.591 to 0.115. Whatever fails at the domain boundary changes the ranking, so it is not calibration in that sense.

2) *What the synthesis boundary does support:* Injecting transients moved the probability in the intended direction and reached nominal significance in the pooled analysis but moved only 50% and 59% of excerpts consistently, at the statistic's floor; we report it as a mean shift with no reliable per-excerpt effect. Envelope compression did not achieve its intended descriptor change and is a failed manipulation rather than a null result.

A specificity diagnostic bounds what may be claimed (Fig. 4c): no intervention moved a single descriptor in isolation, each shifted a correlated family, and no intervention moved its own target descriptor most. Transient injection moves `spec_contrast6_dmean` by 9.0 standard deviations while moving `onset_rate` by 2.0, ranking its target twelfth. Claims are therefore available at the level of a descriptor *dimension* and not of any individual descriptor.

Finally and most importantly, every statement in this section is about model output. No listener heard any edited excerpt. Nothing here establishes that a listener's affective response would move in the same direction, or in any direction.

E. Boundary 3: cross-response-channel

This boundary is crossed by changing what counts as the response while the sound and the listeners stay fixed. It is the boundary an application crosses when it stops asking people how they feel and reads a sensor instead. It also involves two different corpora and two different physiological modalities, which the preprint fused under a single umbrella; we keep them separate.

1) *What was excluded before the question was asked:*

Two analyses were removed on criteria unrelated to what they might have shown. In ds002721 the documented mapping from stimulus trigger code to audio file does not reproduce, and an attempt to recover it by matching emotion profiles did not beat its permutation null ($p = 0.19$); acoustics-to-physiology is therefore *not testable* in that corpus and we do not test it (Supplementary Section S1). In BIRAFFE2 no participant of 94 passed both electrodermal positive controls, so that pipeline is withdrawn (Supplementary Section S2) and supports nothing here.

2) *EEG: ratings carry more stimulus-specific structure than EEG measures do:* Against the four passing controls, stimulus-level reliability was estimated for eight self-report scales and 156 EEG measures on the same 31 listeners and with the same estimator (Fig. 5). Coherence is the strongest EEG family (maximum intraclass correlation 0.092, 95% CI [0.037, 0.142]) and band power the weakest (0.040), against 0.221 [0.155, 0.281] for the best-agreeing self-report scale.

Four qualifications are required, and the preprint stated only the first two. The EEG value is the maximum of 156 measures and is biased upward for that reason; referred to a null built the same way — permuting stimulus labels, recomputing all 156 correlations and taking the largest, 400 times — it carries $p = 0.02$, while Benjamini–Hochberg across the same 156 does not pass it (smallest $q = 0.16$). It is close to the 0.10 this design resolves with 80% power, so the interval is consistent with values from near zero upward. Third, the self-report value is also a maximum of a selection, but over 8 candidates rather than 156, so the two point estimates carry *unequal* selection bias in the direction that favours the EEG side; the comparison is conservative with respect to our conclusion but the asymmetry should be stated. Fourth, the two sets are not computed on identical trials: the EEG measures use 1,240 trials over 307 stimuli and the self-report scales 1,080–1,121 trials over 285–298 stimuli. The preprint's claim of "the same 1,240 trials" for both is incorrect.

What the data support is a comparison and not an absence: whatever stimulus-specific structure these EEG measures carry is less than half of what the listeners' own ratings carry on largely overlapping trials. This is a statement about the component shared across listeners. It is not a claim that the brain does not respond to sound — the onset response in these same recordings is large and highly reliable ($|t| = 8.78$ at 452 ms, $p = 0.0001$). What is weak is the *differentiating* component.

A positive result exists in the literature on these same recordings and we state its relation to ours precisely, because a reader could otherwise take the two as contradictory. Daly et al. [50] report that music-induced emotion can be predicted from brain activity *combined with* acoustic features, at a best correlation of $r = 0.234$ ($p < 0.001$), with either feature type alone performing significantly worse. We do not contradict that. It is a group-level, within-dataset prediction result combining two modalities; ours is a stimulus-level reliability estimate on EEG measures alone, asked in order to bound what a mapping from acoustics could reach. The two are compatible: a combined model can predict an outcome that an individual

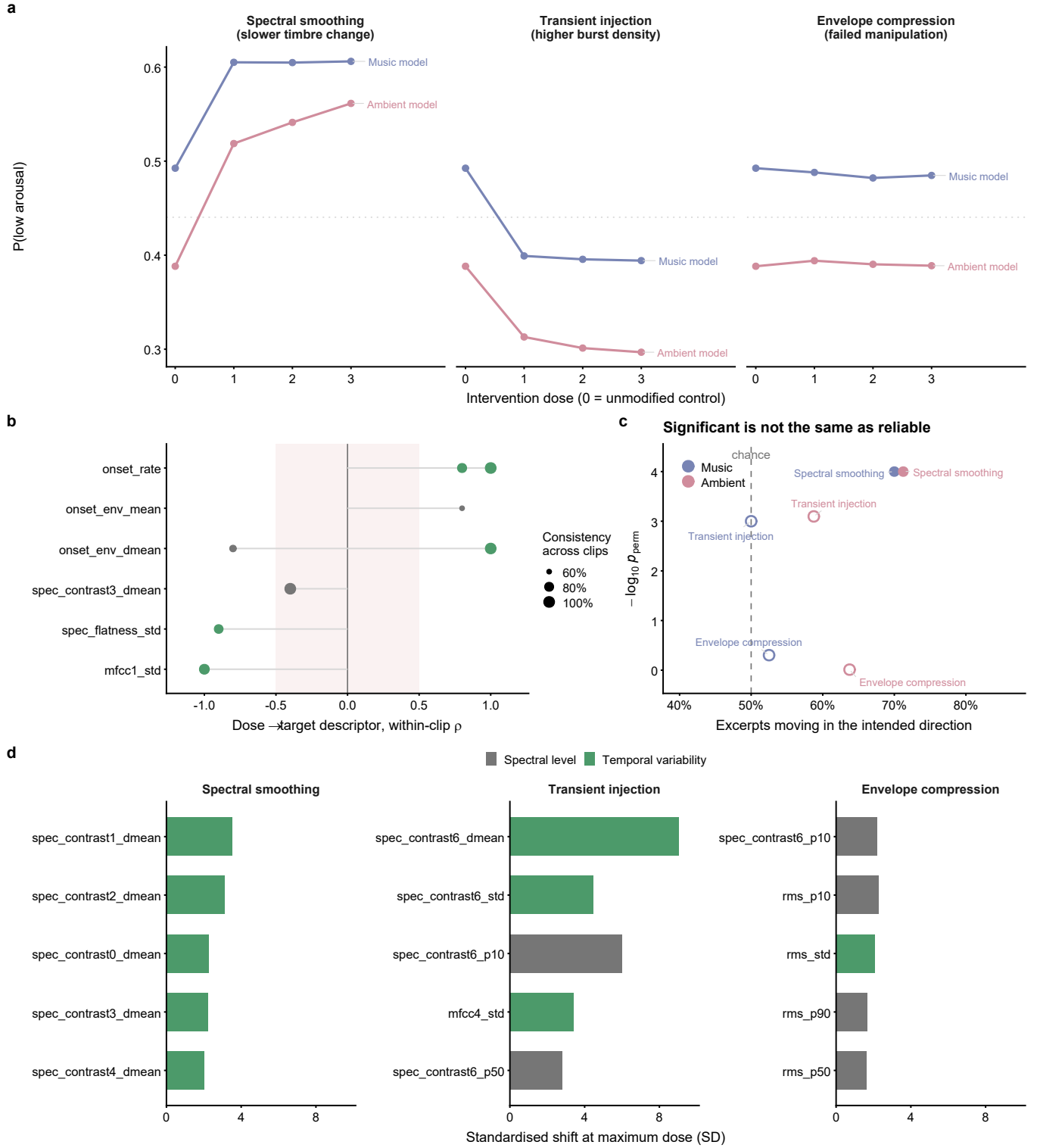


Fig. 4. An acoustic edit moves model output, reliably per excerpt for only one of three interventions and only at the level of a descriptor dimension. **(a)** Dose-response for three interventions on 80 excerpts at four doses; curves are mean predicted probability of the low-arousal class, so a rising curve means the edit made the excerpt read as calmer. **(b)** Manipulation check, run before any dose-response was read. Envelope compression fails it and is carried through as a failed manipulation rather than as a null effect. **(c)** Specificity: the five descriptors moved most by each intervention at maximum dose. No intervention moves its own target descriptor most, which restricts claims to a descriptor *dimension*. **(d)** Significance against per-excerpt reliability. Sign consistency is computed against the observed median's sign, so 50% is its arithmetic floor and not a chance level (null median 0.54). Panels **(a)** and **(d)** pool each model's own-domain and other-domain excerpts; Table IV disaggregates them, and that pooling carried the preprint's withdrawn "calibration, not sign" claim. Every quantity here is model output; no listener heard an edited excerpt.

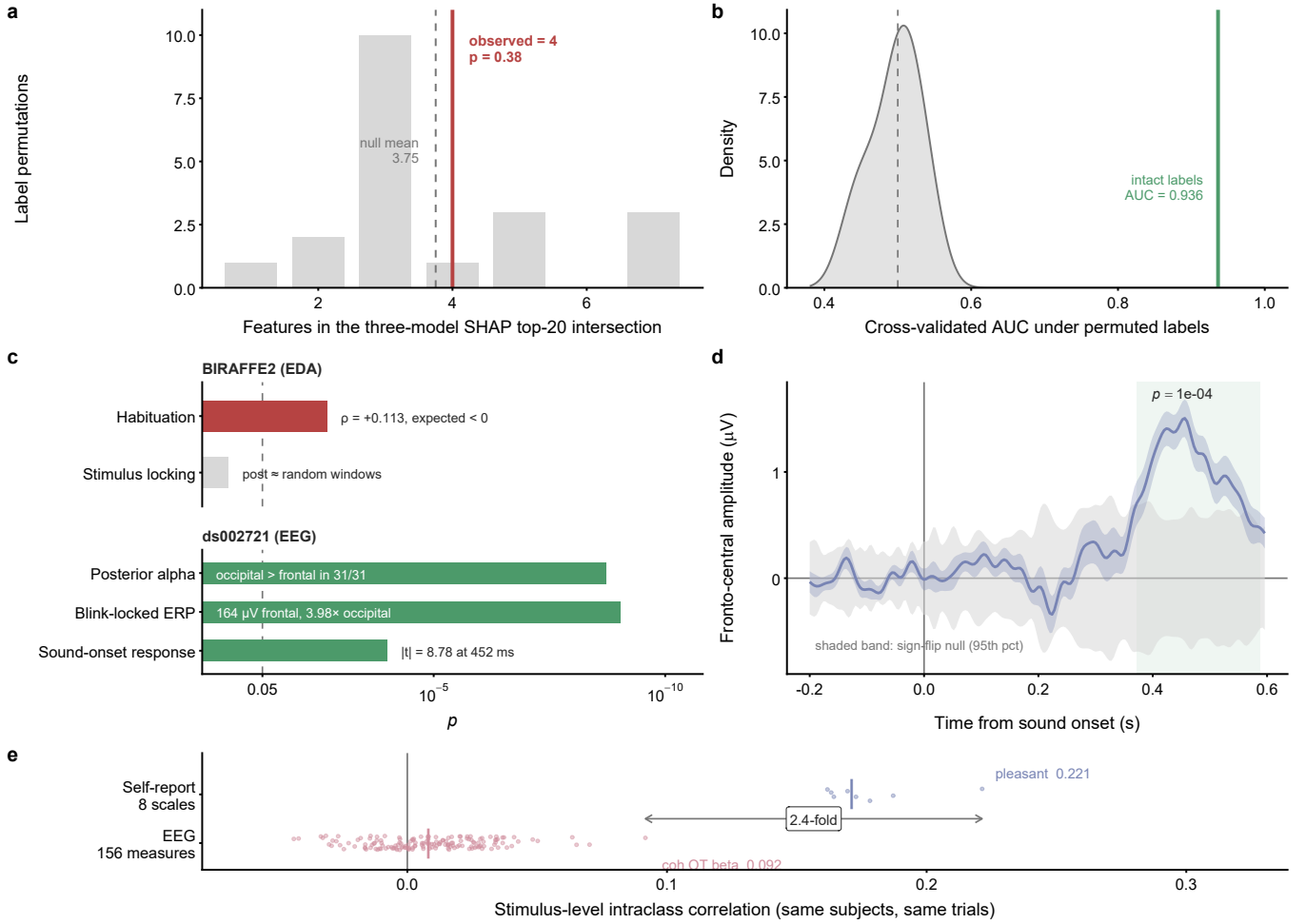


Fig. 5. Reference distributions, positive controls, and the comparison they license. **(a, b)** A cross-model attribution consensus against its label-permutation reference: the observed overlap falls at the null mean ($p = 0.38$) while the intact labels reach an AUC of 0.936, so the statistic cannot separate signal from noise and no claim is built on it. **(c)** Positive controls for two physiological pipelines. BIRAFFE2 fails both, and is withdrawn; ds002721 passes all. **(d)** Grand-average fronto-central response to sound onset ($n = 31$); grey band, 95th percentile of a subject-wise sign-flip null with max-statistic correction; peak $|t| = 8.78$ at 452 ms, $p = 0.0001$. **(e)** Stimulus-level intraclass correlation for eight self-report scales and 156 EEG measures on the same listeners. The EEG value is the maximum of 156 candidates and the self-report value the maximum of 8, so their selection biases are unequal in the direction that favours the EEG side; and the two sets rest on 1,240 and 1,080–1,121 trials respectively, not on identical trials as the preprint stated. Reported as a bound, not an absence.

channel carries only weakly. We also note that the acoustics half of their analysis requires the trigger-to-audio mapping we were unable to reproduce (Supplementary Section S1), which is why we do not attempt the corresponding analysis here rather than reporting it as a failure.

3) *Electrodermal: no acoustic route exceeds its control, and no ceiling can be quoted:* In PMemo, three routes to the song-level electrodermal measures were compared. Acoustic descriptors predict them at $\rho = 0.014$ – 0.070 with p between 0.03 and 0.34; ratings predicted from acoustics reach $\rho = 0.091$ – 0.119 ; and the listeners' own ratings reach $\rho = 0.117$ ($p = 0.004$, valence against skin-conductance response rate). A random projection of the descriptor space reaches the level of the two acoustic routes at its 95th percentile. The defensible statement is therefore: **no acoustic route to these electrodermal measures exceeds an uninformative control, and the listeners' own ratings do, weakly.**

The preprint expressed all three as percentages of an attainable ceiling — 31%, 24.5% and 21.3% — and argued

that three routes stopping within a third of the ceiling made it unlikely that the limit lay in the predictor. Three things are wrong with that argument and we withdraw it in full.

First, the ceiling is not established. It was $\sqrt{r}$ of the measure's stimulus-level reliability, and Section III-D shows that presentation position alone reproduces 114–249% of that reliability. A denominator that a position sham reproduces cannot bound a stimulus-driven predictor.

Second, the percentages selected across denominators. The three figures use two different electrodermal measures whose ceilings differ by 40%, and the largest percentage belongs to the measure with the *lowest* reliability and therefore the lowest ceiling — not to the largest correlation. The highest raw correlation (0.135, valence against phasic mean) yields the *smaller* percentage.

Third, the preprint asserted that the three routes' confidence intervals overlap. No confidence intervals were computed for the acoustic routes. The one interval that does exist, for the rating route, contains the random-projection control.

Finally, the three routes are not independent evidence: route 2 is derived from route 1 and route 3 is the source of route 2. Three nested routes failing alike is weaker than three independent ones, which the preprint did not say.

What remains is a clean negative about the routes tested, on a target whose own stimulus-level reliability this corpus cannot establish. We do not claim that no quantity of target-side data would recover this gap, and we did not measure a target-side learning curve on a physiological outcome. That experiment is the single most informative one this work does not contain.

4) *The individual-level analysis is a design problem, not a null:* Because the intended application is per-person, the same question was asked within listeners on the EEG corpus, with the null drawn from each listener's own label permutations. Of 248 listener $\times$ scale tests, 9 reached $p < 0.05$ against a chance expectation of 12.4 (binomial $p = 0.88$), and none survived correction (Supplementary Fig. S6).

Independent-component removal of ocular artefacts acts here as a sensitivity manipulation check rather than as cleaning: it increased detection of the known effect (individual-level control: 26 $\rightarrow$ 29 listeners passing, $\rho = 0.401 \rightarrow 0.573$) while further reducing the effect of interest (13 $\rightarrow$ 9 nominal detections). A pipeline made measurably more sensitive did not reveal the association, which excludes insufficient signal-to-noise.

It does not exclude insufficient observations. With the roughly 30 rated trials each listener contributed, the design simulation below gives 10% power for a within-person association of $r = 0.40$ and 6% for $r = 0.20$. The individual-level analysis is therefore **uninformative** about within-person associations below approximately $r = 0.5$, and we draw no conclusion from it. The between-listener result does not share this limitation, because it pools 31 listeners at the level of the stimulus.

F. Boundary 4: cross-individual

1) *How many observations per person a design needs:* Synthetic within-person associations of known strength were injected into data generated from the empirical feature covariance of the EEG corpus (Ledoit–Wolf shrinkage [51]) and recovered with the identical pipeline, criterion and permutation test (Fig. 6b). The requirement is steep: 314 observations per person for $r = 0.40$, 564 for $r = 0.30$, 1,364 for $r = 0.20$ and 4,888 for $r = 0.10$, against the ~ 30 this corpus supplies (Table V).

Because that rests on an assumed covariance structure, we measured the same curve directly in ds003690, which supplies ~ 240 auditory events per person with simultaneous EEG, electrocardiogram and pupillometry and carries an effect that is certainly present. Establishing it at the group level first is what makes the individual-level result interpretable: at $p = 0.0002$ for all three channels, an individual in whom it is not detected is one in whom the measurement failed, not one in whom the effect is absent. The cardiac component reproduces the anticipatory deceleration of -0.83 bpm those authors originally reported in this cohort [47].

At the full 240 observations the evoked response is recovered in 72 of 75 listeners from EEG, 62 of 73 from pupil diameter and 47 of 75 from heart rate. At 30 observations — what a

TABLE V
OBSERVATIONS PER PERSON REQUIRED FOR 80% POWER, AND THE POWER AVAILABLE AT THE ~ 30 RATED TRIALS THE EEG CORPUS PROVIDES. SIMULATED ROWS USE INJECTED ASSOCIATIONS OF KNOWN STRENGTH; MEASURED ROWS USE THE AUDITORY EVOKED RESPONSE IN AN INDEPENDENT CORPUS OF 75 LISTENERS.

	Effect	Obs. for 80% power	Power at $n = 30$
Simulated	$r = 0.40$	314	0.10
	$r = 0.30$	564	0.07
	$r = 0.20$	1,364	0.06
	$r = 0.10$	4,888	0.04
Measured	Scalp EEG (N1)	57	0.53
	Pupil diameter	109	0.39
	Heart rate	233	0.23

laboratory session supplies — the same effect is recovered in 53%, 39% and 23% of individuals, while the false-positive rate stays nominal across the whole range (0.03–0.07) (Fig. 6a). An effect plainly visible in the group average ($-5.6 \mu V$ at the N1) is a coin toss within a single person at the observation count these paradigms provide.

Three features bear on how the simulated requirement should be read. It validates the extrapolation: predicting detection at n from an effect size estimated on a disjoint half of the same person's trials matches the observed rate to within 0.03 on average. The requirement is channel-specific in the direction that matters for applied work — heart rate, the signal a consumer device records, needs roughly four times as many observations as scalp EEG (233 against 57). And it is not worse in older listeners; the evoked response is larger in the older group, so fewer observations are needed (48 against 69 for EEG). Both kinds of estimate are lower bounds: the simulation assumes multivariate normality, linearity and within-person stationarity, each of which favours detection, and the measurement uses a deliberately plain pipeline.

One further result from that corpus bears on the choice of channel. In its passive block, where the same tones are presented with no task and no response, the cortical response is undiminished ($-5.7 \mu V$, $p = 0.0002$) but neither the cardiac nor the pupillary response is detectable ($p = 0.30$ and $p = 0.99$). This is not a power failure: had the effect been the size it is in the active blocks, the group test would have detected it with probability 1.00. The scope of the observation is narrow — 250 ms tones, awake listeners, 30 passive trials each — and it does not transfer to music heard over minutes. It does mean that for a passively listening user monitored through heart rate alone, the question of how many observations are needed may be preceded by the question of whether there is a response to count.

2) *Between-person slope heterogeneity, and what it does not establish:* The question an application faces is when fitting a listener their own slope beats giving them the population slope. Table VI reports two answers, and they disagree.

The variance model places break-even at 67 observations per person in urban soundscape ratings, 166 in music ratings, and — over the minority of electrodermal cells in which it is finite at all — 318. Between-person spread exceeds its own

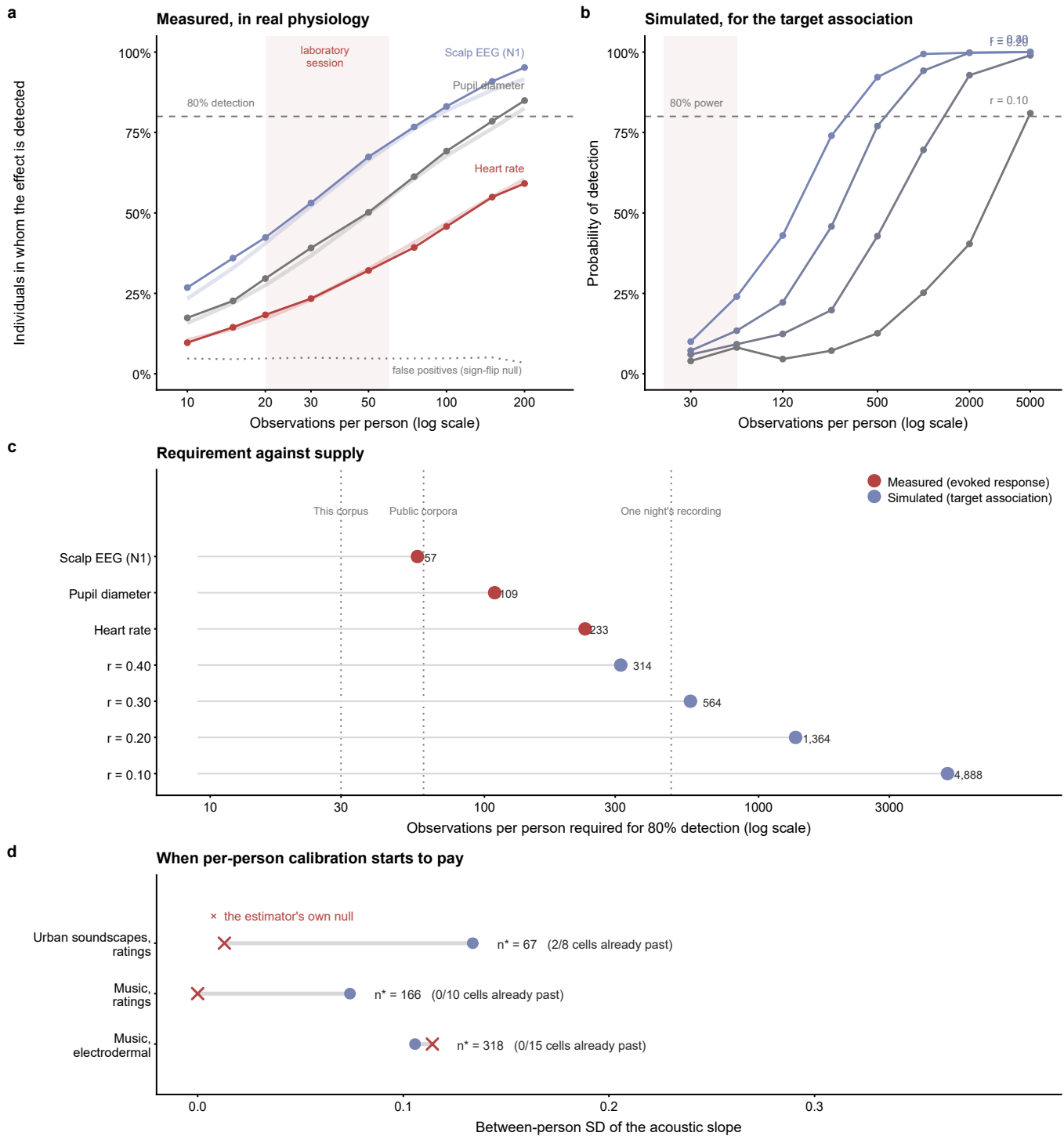


Fig. 6. The individual-level question is a design problem rather than a null result. **(a)** Detection measured in real physiology: in 75 listeners with ~ 240 auditory events each, the proportion in whom the evoked response is detected when only n of that listener's observations are used (40 resamples per point), with each channel's analytic power curve overlaid. The dotted line is the false-positive rate from sign-randomised differences and stays nominal (0.03–0.07) throughout, so the rise is detection. **(b)** Simulated power against observations per person for four true association strengths. **(c)** Observations per person for 80% detection, measured and simulated, against three supply levels. **(d)** Between-person slope spread and its per-cell parametric-bootstrap null in three settings; the null is plotted rather than subtracted because the estimator returns 0.11 at 18 observations per person when the true value is zero. Panel **(d)** is a different quantity from **(a)**–**(c)**, and its three settings differ in corpus, descriptor set and outcome axis together, so they are not compared against one another. Extended description in the Supplementary Information.

per-cell null in 8 of 8 soundscape cells, 10 of 10 music-rating cells, and only 4 of 15 electrodermal cells.

Three properties of the electrodermal row must be stated be-

cause the preprint reported its n^* as though it were comparable to the other two. Ten of the fifteen cells have a corrected spread of exactly zero, so n^* for them is infinite: no budget buys a

quantity that may be zero. The reported 318 is the median over the five cells where n^* is finite, a subset the preprint did not disclose, and one of those five does not exceed its own null. The median over all fifteen cells is infinite, and the median over the four cells that do exceed their null is 264. We give all three and interpret none of them as a price.

3) *The direct test returns a negative gain at every budget available:* The repository behind this paper already contained the out-of-sample test the break-even framing needs, and the preprint did not report it. We report it here because it is the more decisive evidence and it points the other way.

Fitting each listener their own slope and evaluating out of sample against the population slope returns a **negative** gain in 15 of 15 electrodermal cells (-0.109 to -0.072 at 18 observations per person) and in every one of 10 music-rating cells at 18, 30 and 50 observations per person. The gain rises monotonically towards zero with budget: -0.125 to -0.052 at $n = 18$, -0.054 to -0.020 at $n = 30$, -0.028 to -0.002 at $n = 50$, and at $n = 100$ four of ten cells turn positive with the best at $p = 0.054$.

That trajectory is consistent with the variance model's break-even of 166 for this setting, and to that extent it validates the extrapolation's direction. It also settles the applied question in the opposite direction from the preprint's reading: **at every observation count these corpora supply, per-person calibration measurably loses to the population model.**

No out-of-sample test of this kind exists for the soundscape corpus, which is the setting with the lowest n^* and the only one where the model implies break-even has been reached. The two cells said to have crossed break-even are both on the eventfulness axis, and the manipulation check for that axis does not pass in that corpus — the sound-to-masker ratio manipulation moves pleasantness ($p = 4 \times 10^{-6}$) but not eventfulness ($p = 0.38$). The single positive individualisation result in this paper therefore sits on the axis with no passing positive control and with no out-of-sample confirmation. The defensible reading is that per-person calibration in the product-adjacent domain is *worth testing* at a scale existing corpora reach, and that it has not been shown to pay anywhere.

For the same reason we do not compare the three settings against one another. Corpus, descriptor set and outcome axis vary together: the soundscape cells use ISO-532 psychoacoustic descriptors against ISO soundscape axes, the music cells use spectro-temporal and tonal descriptors against valence and arousal, and the electrodermal cells use those same descriptors against electrodermal measures. The preprint concluded from the spread that "choosing the wrong corpus is as expensive as choosing the wrong channel"; with three covarying factors and three settings, that conclusion is not available.

V. DISCUSSION

A. What the evidence supports

Reported as a ledger rather than a verdict, the four boundaries divide by evidence state rather than by kind of failure.

Improvement observed. On the corpus boundary, both the same-domain and the cross-domain gap are substantially recoverable. One hundred target-side labels recover about half

of either gap and 200 recover two-thirds to three-quarters; label-free correlation alignment recovers a quarter of the cross-domain gap, and a valid sham establishes that this gain is target-specific. A targeted representation change recovers a specific deficit — tonal descriptors for perceived happiness — that the spectro-temporal set does not reach.

No improvement observed under the routes tested. Four pretrained representations did not recover the cross-domain gap; on a leak-free test set the best of them improves on 122 hand-specified descriptors by 0.034 in rank correlation, against a gap of 0.49. No acoustic route to song-level electrodermal measures exceeded a random-projection control. No within-person EEG-rating association was detected, and no per-person calibration gain was observed out of sample at any available budget. In each case the statement bounds the routes we ran, and in none of them do we claim that the information required is absent from the input.

Insufficient evidence. Whether the music/environmental gap is a property of the sound domain or of Emo-Soundscapes specifically cannot be decided with one environmental corpus, particularly as that corpus also differs in annotation method. Whether CLAP's advantage reflects pretraining overlap or greater strength on environmental sound cannot be separated. Whether the electrodermal negative reflects the channel, the predictor or the target's measurement cannot be decided in a corpus where presentation position reproduces the target's reliability. And whether per-person calibration pays at 67 observations per person is untested in the one setting where the model says it should.

The single most useful practical consequence is about reporting convention. A zero-shot cross-corpus number is an upper bound on the generalisation problem, not an estimate of it. On the cross-domain boundary, where the unadapted loss is 75% of the target's own within-corpus reference, a hundred target-side labels recover 57% of the gap that loss opens and a label-free second-moment alignment recovers 25% of it with no labels at all. We quote those recovery fractions rather than composing them with the loss into a residual, because the loss is a median over six algorithms while the recovery fractions are measured on ridge — multiplying the two would be exactly the mismatched-denominator error this paper spends a section correcting. A field that reports the first number and not the second systematically overstates how badly its models generalise — and, by the same token, a field that reports an alignment gain without a valid sham overstates how well adaptation works.

B. Two hazards for anyone reusing these corpora

Both are properties of widely used public corpora rather than of our analysis, and both are actionable.

In Emo-Soundscapes, grouped cross-validation on the supplied group identifiers gives no protection against near-duplicate mixture clips spanning folds. The correct grouping is recoverable from the filenames, which encode the constituents; because every mixture shares a constituent with some other mixture, a connected-components grouping collapses all 613 into one group, so the practical options are to evaluate on the 600

TABLE VI

BETWEEN-PERSON SLOPE HETEROGENEITY AND BREAK-EVEN, WITH THE DIRECT OUT-OF-SAMPLE TEST ALONGSIDE. τ IS THE BETWEEN-PERSON STANDARD DEVIATION OF THE ACOUSTIC SLOPE AFTER SUBTRACTION OF A PER-CELL PARAMETRIC-BOOTSTRAP NULL. n^* IS THE MODEL-IMPLIED BREAK-EVEN. THE FINAL COLUMN IS THE MEASURED OUT-OF-SAMPLE GAIN FROM FITTING EACH LISTENER THEIR OWN SLOPE INSTEAD OF THE POPULATION SLOPE, AT THE OBSERVATION COUNT THE CORPUS SUPPLIES. POSITIVE WOULD MEAN PERSONALISATION HELPS.

Setting	Obs. per person	Cells above own null	Median τ	Median n^*	Out-of-sample gain
Soundscape ratings	42	8/8	0.133	67	not measured
Music ratings	44	10/10	0.074	166	−0.054 to −0.020
Music electrodermal	18	4/15	0.000	∞^a	−0.109 to −0.072

^a 318 over the 5 cells with finite n^* ; 264 over the 4 above their own null.

originals or to hold the mixtures out as a block. On our data the difference is 0.10 in the within-corpus reference and a factor of three in cross-domain transfer.

In PMemo, any analysis that treats the song-level electrodermal measures as stimulus-driven — including any that uses their reliability as a ceiling — needs a presentation-position control first.

C. Threats to validity

One environmental corpus. Every cross-domain cell in this paper involves Emo-Soundscapes, so "crossing a domain boundary" and "transferring to this corpus" are the same operation. The corpus differs from the three music corpora in sound domain, in annotation method (pairwise comparison versus direct rating), in clip length and in provenance. We report the result as measured and do not attribute it to the domain alone. Resolving this needs a second environmental affective corpus, which we were unable to obtain in a form permitting this analysis.

Units of replication. The 36 same-domain and 36 cross-domain transfer values are 6 ordered corpus pairs each evaluated with 6 algorithms. The algorithms are not independent replications; the corpus pairs are the units, and there are six of each kind. Dispersion statistics over 36 values should be read accordingly.

Shared material between corpora. The stimuli of the EEG corpus ds002721 are excerpts from the Soundtracks set used as our fourth music corpus. Results involving both are not independent.

Selection and multiplicity. The 84.4% within-corpus figure is the best of six algorithms; the EEG intraclass correlation is the maximum of 156 measures and the self-report value the maximum of 8, an asymmetry that favours the EEG side. No multiplicity control is applied across the 33 per-person calibration cells. The per-cell nulls for those cells rest on 60 bootstrap replicates, so Monte-Carlo error on the threshold is of the same order as the margin by which the marginal cells clear it, and the slopes are estimated in two stages where a cross-classified hierarchical model would propagate the first stage's uncertainty into the second.

Construct scope. This paper moves between model outputs, group-averaged perceived-emotion ratings, individual self-report, EEG measures, electrodermal measures, heart rate and pupil diameter. These are not interchangeable and a result on one does not stand in for another. In particular, perceived

arousal in a rating corpus is not sleep onset, sleep continuity or any clinical endpoint; no analysis here relates a naturalistic soundscape to a peripheral physiological measure; and nothing in this work bears on the efficacy of audio for sleep. Cardiac and pupillary signals enter only through the design-requirement measurement, where the stimulus is a 250 ms tone presented to awake listeners performing a task.

No prior specification. As stated in the Methods, we can produce no traceable artifact predating the analyses for any criterion, and nothing here is preregistered.

D. What would settle the open questions

Three experiments, in order of how much they would change.

A target-side learning curve on a physiological outcome. Everything we can say about that boundary concerns the routes we tried; only a budget curve distinguishes a gap that observations close from one they do not, and none exists.

A second environmental affective corpus, annotated by the same method as the music corpora, outside the pretraining data of the representations under test. That single addition separates domain from dataset, annotation method from sound, and pretraining overlap from representational strength.

Within-person longitudinal data with varied stimuli. The requirement measured here is 57 to 233 observations per person depending on channel, and the corpora available supply 18 to 44. The shortfall is in duration and repetition, not in participants: recruiting more people does not provide observations within a person. Observation count and stimulus richness also trade against each other inside a fixed session — the corpus that supplies 240 events per person does so with four pure tones — and it is the combination that no session-length design provides. A search of OpenNeuro (1,833 datasets) and PhysioNet (426) returned no corpus pairing repeated controlled auditory stimulation with a state readout at the required density, and the standard affective-physiology corpora distributed elsewhere supply 8 to 40 stimuli per participant. That claim is bounded by the search rather than absolute.

VI. CONCLUSION

Across four boundaries, the generalisation losses an affective audio model suffers are large when measured zero-shot and substantially smaller when a modest adaptation budget is quoted alongside them. On the cross-domain corpus boundary the unadapted loss is three-quarters of the target's own within-corpus reference, and of the gap that opens, a hundred target

labels recover 57% and a label-free second-moment alignment recovers a quarter with no labels at all — a gain a sham control supports as target-specific on the three pairs where that control is valid. On the response-channel and individual-listener boundaries we found no route that recovered the gap, and we report that as a bound on the routes tested rather than as a property of the input, because the experiments that would distinguish the two — a target-side budget curve on a physiological outcome, and an out-of-sample personalisation test in the product-adjacent corpus — were not available to us. Two of the corpora we used carry structural hazards that inflate within-corpus performance and distort any ceiling computed from them, and both are recoverable from the distributed metadata. We would rather report a measured price list with three acknowledged gaps in it than a framework that the same measurements contradict.

VII. DECLARATIONS

Ethics. This work is a secondary analysis of de-identified data released for public reuse. No new data were collected from human participants and no attempt was made to re-identify any individual. Approval provenance differs across the nine corpora and is stated per corpus in Supplementary Table S13: where the source publication reports an approving body and protocol identifier we cite it; where it does not, we state that no approval statement is available in the source rather than implying one exists.

Competing interests. The authors are engaged in the development of a commercial sleep-audio product. Nothing in this paper bears on the efficacy of audio for sleep. Four of the corpora used here could not enter such a product in any case, for two different reasons: ESC-50 and ARAUS carry non-commercial terms, and the audio of PMemo and of Soundtracks carries no redistribution right — the latter because the Soundtracks deposit's own CC BY 4.0 licence does not convey rights in the commercially released recordings it excerpts. One of the pretrained representations, MERT, is likewise non-commercial.

Use of generative AI. Claude (Anthropic) was used under author direction to generate and revise analysis code, assist with pre-submission consistency checks, execute the reanalyses reported in Sections III-D, IV-B, and IV-D, and draft and revise text throughout the manuscript and supplementary material. Numerical analyses and figures were produced by executing the analysis and plotting scripts described in the repository. The authors reviewed the AI-assisted text and code, checked the reported results against the analysis outputs, and verified the references. Generative AI was not used to fabricate participant records or empirical observations; simulations and audio transformations are described in the Methods. The authors take full responsibility for the methods, results, interpretations, and final manuscript.

Data availability. All analyses use publicly available data. Accession identifiers, access dates and licence terms are listed in Supplementary Table S14. No new data were generated. Derived tables underlying every figure panel are provided as supplementary source data.

Code availability. Analysis code, run records and figure-generation scripts are at <https://github.com/AuraJZ/affective-audio-boundaries>. Each reported value has a run record identifying the script, the random seed and the parameters. The limits of that record are material and are stated rather than glossed: all run records were written from a working tree with uncommitted changes and the repository head is older than most of the analyses, so the recorded commit identifies a starting point and not the state that produced the number; input hashes are present for 38 of 199 records and output hashes for none; there is no automated test suite. The analyses are reproducible from the scripts and the public corpora at the parameters recorded, and are not reproducible bit-for-bit from a pinned commit. We do not claim the stronger guarantee.

Author contributions. **J.Z.:** conceptualisation; methodology; software; validation; formal analysis; investigation; data curation; visualisation; writing — original draft; writing — review and editing. **X.Y.:** conceptualisation; resources; validation; writing — review and editing; project administration.

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